Example 11.28

Evaluate : $\int \frac{1}{\sqrt{x+1}+\sqrt{x}}\,dx.$

Solution

$$\int \frac{1}{\sqrt{x+1}+\sqrt{x}}\,dx = \int \frac{1}{\sqrt{x+1}+\sqrt{x}}\left[\frac{\sqrt{x+1}-\sqrt{x}}{\sqrt{x+1}-\sqrt{x}}\right]dx$$

$$= \int \frac{\sqrt{x+1}-\sqrt{x}}{\left(\sqrt{x+1}^{2}\right)-\left(\sqrt{x}\right)^{2}}\,dx$$

$$= \int \frac{\sqrt{x+1}-\sqrt{x}}{x+1-x}\,dx = \int\left(\sqrt{x+1}-\sqrt{x}\right)dx$$

$$= \int \sqrt{x+1}\,dx - \int \sqrt{x}\,dx = \int (x+1)^{\frac{1}{2}}\,dx - \int x^{\frac{1}{2}}\,dx$$

$$= \frac{(x+1)^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c$$

$$= \frac{2}{3}\left[(x+1)^{\frac{3}{2}} - x^{\frac{3}{2}}\right] + c.$$

11.7.2 Decomposition by Partial Fractions

One of the important methods to evaluate integration is partial fractions. If the integrand is in the form of an algebraic fraction and the integral cannot be evaluated by simple methods, then the fraction need to be expressed in **partial fractions** before integration takes place. We will assume that we have a rational function $\frac{p(x)}{q(x)}$, $\left(q(x) \neq 0\right)$ in which degree of $p(x)$ < degree of $q(x)$. If this is not the case, we can always perform long division.

Example 11.29

Evaluate : (i) $\int \frac{3x+7}{x^2-3x+2}\,dx$ (ii) $\int \frac{x+3}{(x+2)^2(x+1)}\,dx.$

Solution

(i) $$\int \frac{3x+7}{x^2-3x+2}\,dx = \int \frac{13}{x-2}\,dx - \int \frac{10}{x-1}\,dx$$ Resolving into partial fractions

$$= 13\log|x-2| - 10\log|x-1| + c$$

(ii) $$\int \frac{x+3}{(x+2)^2(x+1)}\,dx = \int \frac{-2}{x+2}\,dx - \int \frac{1}{(x+2)^2}\,dx + \int \frac{2}{x+1}\,dx$$ Resolving into partial fractions

$$= -2\int \frac{1}{x+2}\,dx - \int \frac{1}{(x+2)^2}\,dx + 2\int \frac{1}{x+1}\,dx$$

$$= -2\log|x+2| - \int (x+2)^{-2}\,dx + 2\log|x+1| + c$$

$$= -2\log|x+2| + \frac{1}{x+2} + 2\log|x+1| + c.$$

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